UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 7D – Predetermined Motion Time Systems I

EIN3241: Human Factors and Ergonomics
Spring 2026

April 6, 2026

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Admin Issues

- Friday, April 10: Peer Evaluation 1
- Wednesday, April 15: Last Quiz (Quiz 6)
  - Time Studies, Work Sampling, and PMTS
- April 17, 20, and 22
  - Project Presentations
  - Mandatory Attendance
- April 25: Project Report
- April 26: Peer Evaluation 2
- April 28: Final Exam
  - Cumulative
  - 8:00-10:00 PM in FLG260

Example 2

- Consider the following results of a pilot work sampling study. With an operator rating of 110% and an allowance of 15%, what is the standard time per unit?

Information	Data
Total working day	480 min
Number of units drilled	420
Working fraction	85%

Work Measurements

- How to calculate standard time
  - Work sampling (Activity Sampling)
  - Time studies
  - **<u>Predetermined time systems (PMTS)</u>**
    - Most popular form of finding standard time today
    - Very useful in large and complex systems
    - Gives you the target standard time
  - Standard data
  - Time formulas
  - Task analysis

Predetermined motion time system (PMTS)

- Human terms: Break down a job into its most basic parts and then add the time for each part based on predetermined time estimates
- PMTS is a work measurement technique whereby times established for basic human motions are used to build up the time for a job at a defined level of performance.
- PMTS also called **predetermined time system (PTS)**, is a database of basic motion elements and their associated normal time values, together with a set of procedures for applying the data to analyze *manual tasks* and establish standard times for the tasks.
- There are some assumption for PMTS which are independent and additive; that is, each element does not affect what happens before or after it.

History of PTS

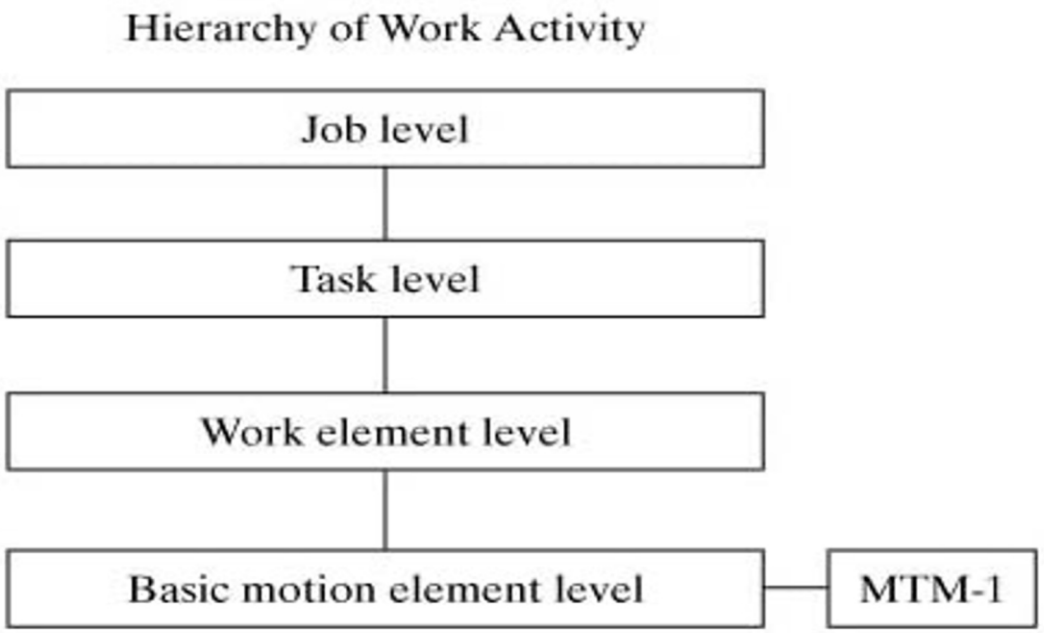
- First PTS was developed by Maynard, Schwab and Stegemerten when consulting for Westinghouse Brake and Signal Corporation in the 1940s
  - Studied videos frame by frame
  - Established times for reach, grasp, move, etc.
- PTS today:
  - Main advantages
    - Can be done before the job is set up
    - Does not require any workers, experiments, or subjective ratings
  - Main disadvantages
    - Can be tedious to do
    - Requires training and certification to be done correctly
    - May not always be entirely realistic

Methods Time Measurement (MTM)

- Earliest PTS established (1948)
- MTM is the result of frame-by-frame analyses of motion picture films
- Has many different forms
  - MTM-1
  - MTM-2
  - MTM-3
  - MTM-V (metal cutting operations)
  - MTM-C (banking and insurance)
  - MTM-M (operator work using a microscope)
  - Others

MTM-1

- Gives time values for the fundamental motions
  - Reach
  - Move
  - Turn
  - Apply pressure
  - Grasp
  - Position
  - Disengage
  - Release
  - Eye travel time and focus
  - Body, leg, and foot motions



MTM–1

- MTM breaks down all basic motions into units of **TMU (time-measurement unit)**
  - 1 TMU = 0.00001 hr
  - 1 TMU = 0.0006 min
  - 1 TMU = 0.036 sec
  - 1 hour = 100,000 TMU
  - 1 min = 1667 TMU
  - 1 sec = 27.8 TMU
- Times are for a trained (normal) worker working at a normal pace.
- No allowances are included in the times.

MTM–1

- Steps for determining standard time using MTM-1
  1. Break down the left- and right-hand motions into steps
  2. Break down the steps into sequences of the basic motions
  3. Assign the appropriate symbol for each basic motion
  4. Find the equivalent TMUs from the MTM-1 tables
  5. Remove the non-limiting values
  6. Calculate standard time

Break down job into steps

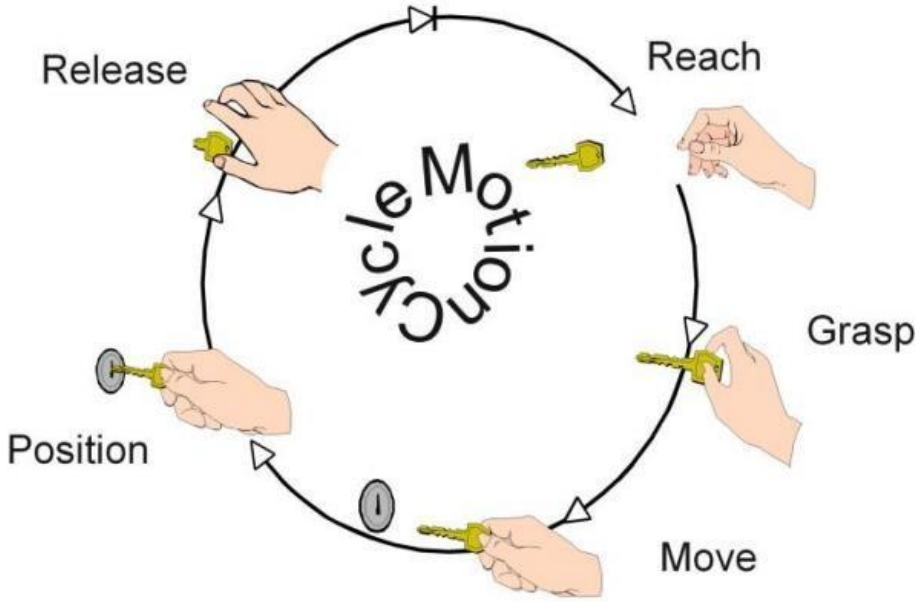
- Document the job
  - Even though job might not be set up yet, must know the standard method
  - List the materials, equipment, purpose, etc.
- List the steps needed in order, typically in a table
  - Indicate what the left and right hands are doing
  - Can indicate if any steps need to be repeated using a frequency column (e.g., x2)
  - Steps that do not always occur are usually studied separately (linear form is preferred)
- Replace page in a 3-ring binder (left hand)
  - Get binder from shelf to table
  - Open binder
  - Locate correct page
  - Replace page
  - Close binder
  - Put aside

MTM-1

- Steps for determining standard time using MTM-1
  1. Break down the left- and right-hand motions into steps
  2. Break down the steps into sequences of the basic motions
     - Indicate the frequency of basic motions if applicable (column F)
  3. Assign the appropriate symbol for each basic motion
  4. Find the equivalent TMUs from the MTM-1 tables
  5. Remove the non-limiting values
  6. Calculate standard time

MTM-1

- Fundamental or basic motions:
  - Reach
  - Move
  - Turn
  - Apply pressure
  - Grasp
  - Position
  - Disengage
  - Release
  - Eye travel time and focus
  - Body, leg, and foot motions



Break down into basic motions

1. Get binder from shelf to table
    1. Reach to binder
    2. Grasp binder
    3. Move binder to desk
    4. Release

Left-Hand Description	Right-Hand Description
1. GET BINDER	
Reach to binder	
Grasp binder	
Move to desk	
Release	

MTM–1

- Steps for determining standard time using MTM-1
  1. Break down the left- and right-hand motions into steps
  2. Break down the steps into sequences of the basic motions
  3. Assign the appropriate symbol for each basic motion
  4. Find the equivalent TMUs from the MTM-1 tables
  5. Remove the non-limiting values
  6. Calculate standard time

MTM-1

- For each of the fundamental or basic motions, assign a symbol to indicate what type it is and how much time it will take
  - Reach
  - Move
  - Turn
  - Apply pressure
  - Grasp
  - Position
  - Disengage
  - Release
  - Eye travel time and focus
  - Body, leg, and foot motions

Reach

- Reach involves moving the hand or fingers to a definite destination or to a general location. It depends on:
  - Distance
  - Type of reach
    A. Reach to object in fixed location, or to object in other hand, or to object on which other hand rests.
    B. Reach to single object in location that may vary slightly from cycle to cycle.
    C. Reach to object jumbled with other objects, requiring search and select as well.
    D. Reach to a very small object, or an object for which accurate grasp is required.
    E. Reach to indefinite location to get hand in position for body balance, or for next motion, or out of the way.
- Example: Reaching an object jumbles with others objects in group from 1 inch distance.
- The code:
  - **R 1 C** takes 3.6 TMU = 0.130 sec
  - **R: Motion element symbol: Reach**
  - **1: Distance: 1 inch**
  - **C: Case description: Reaching an object jumbles with others objects in group**

Reach

Table 13.1 Summary of MTM-1 Data

Table I—Reach—R							
Distance Moved (in)	Time (TMU) A	B	C or D	E	Hand in Motion A	B	Case and Description
½ or less	2.0	2.0	2.0	2.0	1.6	1.6	A Reach to object in fixed location, or to object in other hand or on which other hand rests.
1	2.5	2.5	3.6	2.4	2.3	2.3	
2	4.0	4.0	5.9	3.8	3.5	2.7	
3	5.3	5.3	7.3	5.3	4.5	3.6	
4	6.1	6.4	8.4	6.8	4.9	4.3	B Reach to single object in location which may vary slightly from cycle to cycle.
5	6.5	7.8	9.4	7.4	5.3	5.0	
6	7.0	8.6	10.1	8.0	5.7	5.7	
7	7.4	9.3	10.8	8.7	6.1	6.5	C Reach to object jumbled with other objects in a group so that search and select occur.
8	7.9	10.1	11.5	9.3	6.5	7.2	
9	8.3	10.8	12.2	9.9	6.9	7.9	
10	8.7	11.5	12.9	10.5	7.3	8.6	D Reach to a very small object or where accurate grasp is required.
12	9.6	12.9	14.2	11.8	8.1	10.1	
14	10.5	14.4	15.6	13.0	8.9	11.5	
16	11.4	15.8	17.0	14.2	9.7	12.9	E Reach to indefinite location to get hand in position for body balance or next motion or out of way.
18	12.3	17.2	18.4	15.5	10.5	14.4	
20	13.1	18.6	19.8	16.7	11.3	15.8	
22	14.0	20.1	21.2	18.0	12.1	17.3	
24	14.9	21.5	22.5	19.2	12.9	18.8	
26	15.8	22.9	23.9	20.4	13.7	20.2	
28	16.7	24.4	25.3	21.7	14.5	21.7	
30	17.5	25.8	26.7	22.9	15.3	23.2	

R5B

Add 0.8 TMU per inch for each additional inch

Move

- Move involves holding and changing the location of an object. It depends on:
  - Distance
  - Weight of object
  - Type of move:
    A. Move object to other hand, or against stop
    B. Move object to approximate or indefinite location.
    C. Move object to exact location.
- Move involves holding and changing the location of an object. It depends on:
  - There is also a static and dynamic factor that is accounted for:
  - Static factor added when hand has just grasped the object in preparation for move (when starting from rest)
  - Dynamic factor accounts for resistance when moving

Move

M12A10 (Weight of 10 pounds)

Table II—Move—M								
	Time (TMU)				Wt. Allowance			
Distance Moved (in)	A	B	C	Hand in Motion B	Wt. (lb) up to	Factor	Constant (TMU)	Case and Description
1/2 or less	2.0	2.0	2.0	1.7	2.5	0	0	A Move object to other hand or against stop.
1	2.5	2.9	3.4	2.3				
2	3.6	4.6	5.2	2.9	7.5	1.06	2.2	
3	4.9	5.7	6.7	3.6				
4	6.1	6.9	8.0	4.3	12.5	1.11	3.9	
5	7.3	8.0	9.2	5.0				
6	8.1	8.9	10.3	5.7	17.5	1.17	5.6	
7	8.9	9.7	11.1	6.5				
8	9.7	10.6	11.8	7.2				B Move object to approximate or indefinite location.
9	10.5	11.5	12.7	7.9	22.5	1.22	7.4	
10	11.3	12.2	13.5	8.6				
12	12.9	13.4	15.2	10.0	27.5	1.28	9.1	
14	14.4	14.6	16.9	11.4				
16	16.0	15.8	18.7	12.8	32.5	1.33	10.8	
18	17.6	17.0	20.4	14.2				
20	19.2	18.2	22.1	15.6	37.5	1.39	12.5	
22	20.8	19.4	23.8	17.0				
24	22.4	20.6	25.5	18.4	42.5	1.44	14.3	C Move object to exact location.
26	24.0	21.8	27.3	19.8				
28	25.5	23.1	29.0	21.2	47.5	1.50	16.0	
30	27.1	24.3	30.7	22.7				

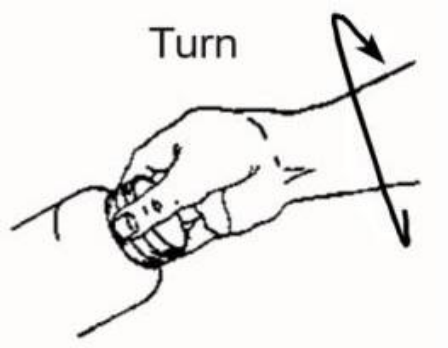
Use the values that are higher up

Divide weight by 2 if both hands moving the object

Add 0.8 TMU per inch for each additional inch

Turn

- Turn is a movement that rotates the hand and wrist. It depends on
  - The number of degrees turned
  - The weight of the object or the resistance against which the turn is made
    - Small (S)
    - Medium (M)
    - Large (L)



T30S

T45M

Table III—Turn and Apply Pressure—T and AP											
	Time TMU for Degrees Turned										
Weight	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°
Small—0 to 2 lb	2.8	3.5	4.1	4.8	5.4	6.1	6.8	7.4	8.1	8.7	9.4
Medium—2.1 to 10 lb	4.4	5.5	6.5	7.5	8.5	9.6	10.6	11.6	12.7	13.7	14.8
Large—10.1 to 35 lb	8.4	10.5	12.3	14.4	16.2	18.3	20.4	22.2	24.3	26.1	28.2

Apply Pressure Case A—10.6 TMU. Apply Pressure Case B—16.2 TMU

Apply pressure

- Apply pressure is the application of force without resultant movement
  - Case A: application of pressure
  - Case B: application of pressure with higher precision
    - Use when you need to hold something carefully in place
    - Use when there is a position before applying pressure

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Medium—2.1 to 10 lb	4.4	5.5	6.5	7.5	8.5	9.6	10.6	11.6	12.7	13.7	14.8
Large—10.1 to 35 lb	8.4	10.5	12.3	14.4	16.2	18.3	20.4	22.2	24.3	26.1	28.2

Apply Pressure Case A—10.6 TMU. Apply Pressure Case B—16.2 TMU

APA

APB

Grasp

- Gain control of an object; often followed by Move.
  1. Pick up grasp: grasp an individual object
     A. By itself (most common)
     B. Small object lying against the surface (like a small paper)
     C. Cylindrical objects (like pens or markers next to each other)
  2. Regrasp: change grasp without relinquishing control; object stays in your hand
  3. Transfer: control transferred from one hand to the other
  4. Select: objects jumbled with each other such that search and select often occur
     A. Large
     B. Medium
     C. Small
  5. Contact: contact or sliding grasp (like pressing a button); your fingers to do not really hold on to anything

Grasp

G1C2

Table IV—Grasp—G		
Case	**Time (TMU)**	**Description**
1A	2.0	**Pick Up Grasp**—Small, medium or large object by itself, easily grasped.
1B	3.5	Very small object or object lying close against a flat surface.
1C1	7.3	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter larger than $^1/_2''$.
1C2	8.7	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter $^1/_4''$ to $^1/_2''$.
1C3	10.8	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter less than $^1/_4''$.
2	5.6	**Regrasp.**
3	5.6	**Transfer Grasp.**
4A	7.3	Object jumbled with other objects so search and select occur. Larger than $1'' \times 1'' \times 1''$.
4B	9.1	Object jumbled with other objects so search and select occur. $^1/_4'' \times {}^1/_4'' \times {}^1/_8''$ to $1'' \times 1'' \times 1''$.
4C	12.9	Object jumbled with other objects so search and select occur. Smaller than $^1/_4'' \times {}^1/_4'' \times {}^1/_8''$.
5	0	Contact, sliding or hook grasp.

Position

- Align, orient, and engage one object with another object. Depends on:
  - Class of fit (loose, close, exact):
    - Loose: like a cup to its saucer
    - Close: like a marker cap to a marker
    - Exact: like a key to its keyhole
  - Symmetry (symmetrical, semi-symmetrical, non- symmetrical)
    - Symmetrical: infinity of ways to insert the object (like a pen inside a cup)
    - Semi-symmetrical: finite ways to insert the object (like a paper into a rectangular cardboard box)
    - Non-symmetrical: one and only one way to insert the object (like a key inside a keyhole)
  - Ease of handling: how easy the object is to hold (slippery, dangerous, sharp…)

P1SE

Table V—Position*—P

Class of Fit		Symmetry	Easy to Handle	Difficult to Handle
1—Loose	No pressure required.	S	5.6	11.2
		SS	9.1	14.7
		NS	10.4	16.0
2—Close	Light pressure required.	S	16.2	21.8
		SS	19.7	25.3
		NS	21.0	26.6
3—Exact	Heavy pressure required.	S	43.0	48.6
		SS	46.5	52.1
		NS	47.8	53.4

*Distance moved to engage—1″ or less.

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LEADING THE CHARGE, CHARGING AHEAD